

“Intelligent AI-Based Smart Dairy Farm Management System”

CO-2 ASSESSMENT TOOL-2

Architecture Design Assignment

Submitted in the partial fulfilment for the course of

CSA1017 - SOFTWARE ENGINEERING

for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Under the Supervision of

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SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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July 2026

RUBRICS FOR CALCULATION

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	Mark
System Requirement Analysis	/5
Selection of Software Architecture	/5
Architecture Diagram and Component Design	/5
Component Interaction and Quality Attribute Analysis	/5
Architectural Justification and Technical Presentation	/5
TOTAL	/25

PROBLEM STATEMENT:-

Traditional dairy farm management relies heavily on manual observation, making it difficult to detect animal health issues, optimize feeding schedules, and maintain production efficiency. An AI- powered dairy farm management system should monitor cattle health using wearable sensors, predict diseases, optimize feeding schedules, track milk production, and provide real-time farm analytics. The platform should improve livestock welfare while increasing dairy productivity.



1.System Requirement Analysis

1.1 Functional Requirements

Requirement ID	Functional Requirement
FR1	User Registration and Login
FR2	Monitor cattle health using wearable IoT sensors
FR3	Predict diseases using AI models
FR4	Schedule and monitor cattle feeding
FR5	Track daily milk production
FR6	Monitor temperature and humidity in the farm
FR7	Send emergency alerts for abnormal health conditions
FR8	Generate reports and analytics
FR9	Manage cattle records
FR10	Backup data to cloud storage

1.2 Non-Functional Requirements

Requirement	Description
Performance	Real-time monitoring with low response time
Security	Secure login, encrypted data, role-based access
Scalability	Supports thousands of cattle and multiple farms
Reliability	Continuous monitoring with minimal downtime
Availability	System available 24×7
Maintainability	Easy updates and modular design
Usability	User-friendly interface for farmers
Portability	Accessible through web and mobile applications

1.4 Quality Attributes

Scalability :-

The system supports increasing numbers of cattle, sensors, and users without reducing performance.

Security :-

Uses authentication, authorization, encrypted communication (HTTPS), and secure cloud storage.

Performance :-

Processes IoT sensor data quickly and provides real-time analytics.

Reliability :-

Automatically detects failures, stores backups, and continues monitoring.

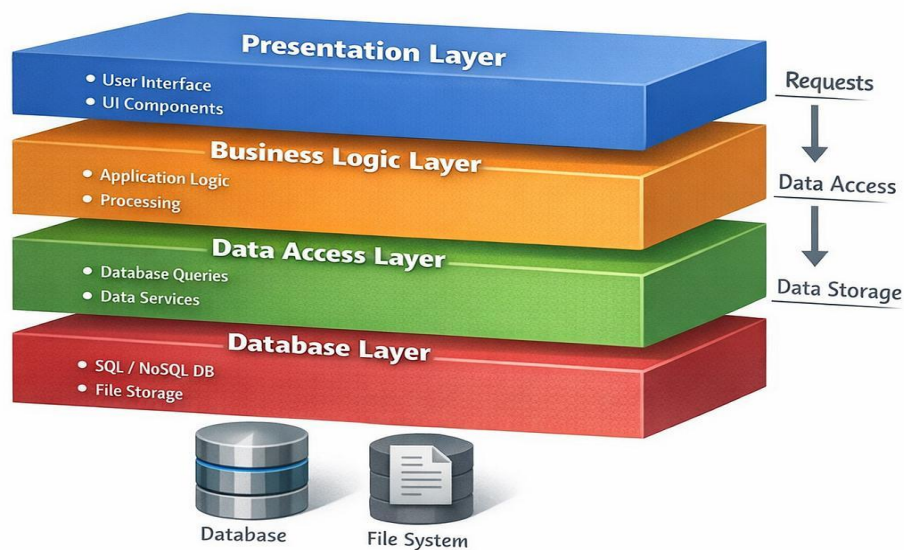
Availability :-

Cloud deployment ensures continuous access and disaster recovery.

Maintainability :-

Layered architecture allows easy modification and future enhancements.

2. Selection of Software Architecture



Selected Architecture

Layered Architecture

Why Layered Architecture?

The Layered Architecture divides the system into separate layers, making development, testing, maintenance, and future upgrades easier.

Layers include:

1. Presentation Layer
2. Application Layer
3. Business Logic Layer
4. AI & Analytics Layer
5. Database Layer
6. IoT Device Layer

Advantages

- Easy maintenance
- High scalability
- Better security
- Modular design
- Easy testing
- Supports future enhancements
- Simplifies debugging
- Promotes code reuse

Architecture Layers

Presentation Layer

- Farmer Dashboard
- Mobile App
- Web Portal

Application Layer

- User Authentication
- API Services
- Notification Service

Business Logic Layer

- Health Monitoring
- Feeding Management
- Milk Production Management
- Report Generation

AI & Analytics Layer

- Disease Prediction
- Feeding Optimization
- Productivity Analysis

Database Layer

Stores

- Farmer Information
- Cattle Records
- Sensor Data
- Milk Production
- Reports

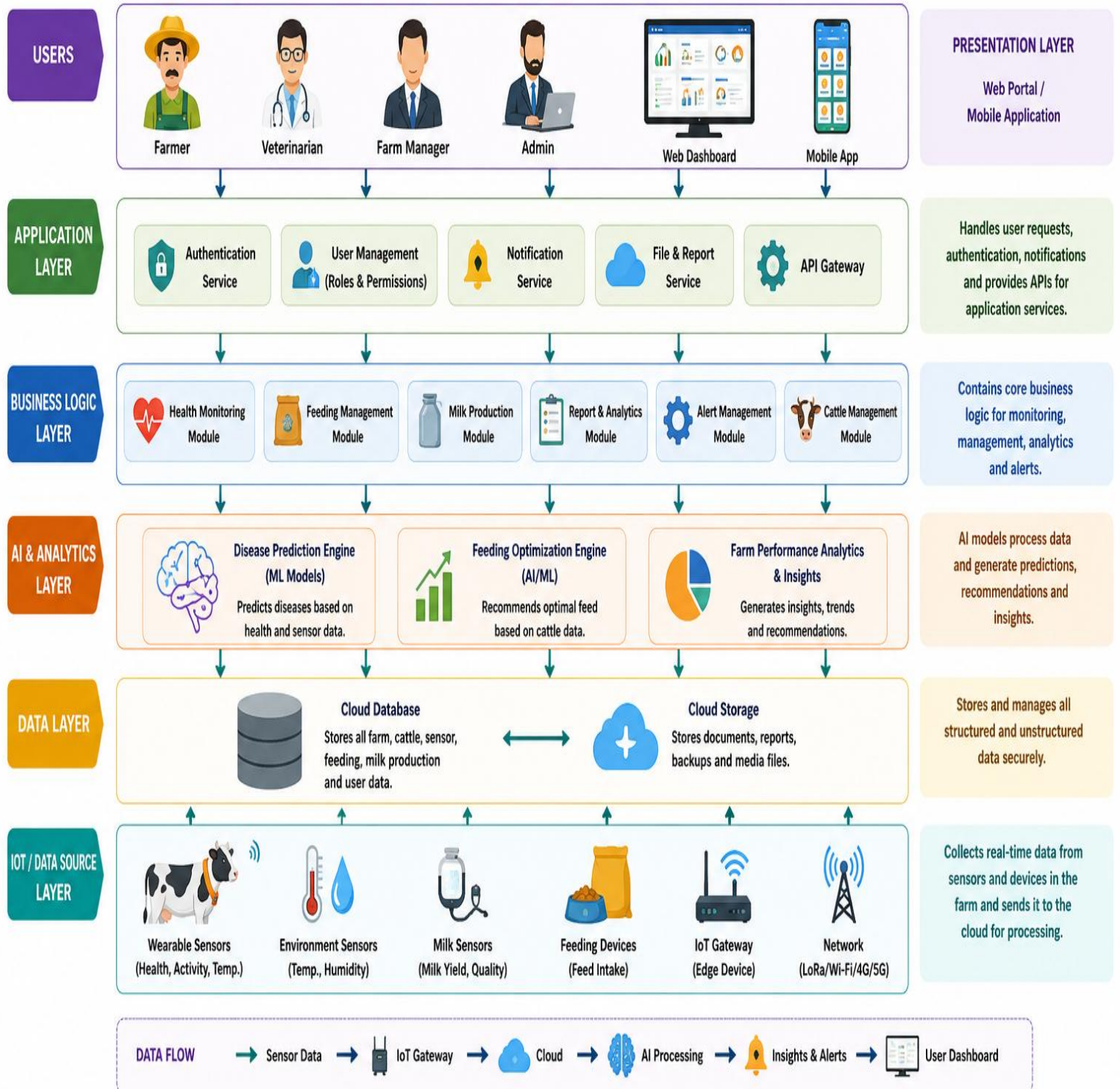
IoT Layer

Includes

- Wearable Sensors
- Temperature Sensors
- Humidity Sensors
- Milk Sensors

3. Software Architecture Diagram

INTELLIGENT AI-BASED SMART DAIRY FARM MANAGEMENT SYSTEM ARCHITECTURE DIAGRAM



4. Components and Their Interactions :-

4.1 Presentation Layer

Responsibilities

- Farmer login
- Dashboard
- View reports
- Receive alerts

4.2 Authentication Module

Responsibilities

- User Registration
- Login
- Role Management
- Secure Access



4.3 Health Monitoring Module

Responsibilities

- Collect wearable sensor data
- Monitor cattle body temperature
- Heart rate monitoring
- Activity monitoring

4.4 Feeding Management Module

Responsibilities

- Schedule feeding
- Monitor feed intake
- Optimize nutrition using AI

4.5 Milk Production Module

Responsibilities

- Daily milk recording
- Milk quality tracking
- Production analysis

4.6 AI Prediction Module

Responsibilities

- Disease Prediction
- Feeding Optimization
- Health Risk Analysis

4.7 Reporting Module

Responsibilities

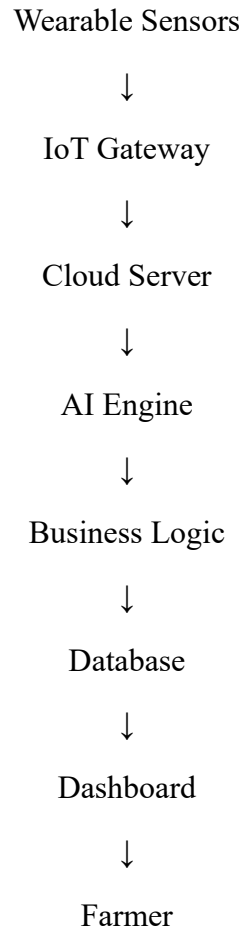
- Farm reports
- Milk reports
- Disease reports
- Feeding reports

4.8 Database

Stores

- Farmer Details
- Cattle Details
- Sensor Data
- Feeding Records
- Milk Records
- AI Predictions

Component Interaction:-



5. Quality Attribute Analysis :-

Scalability

- Supports multiple farms
- Supports thousands of IoT devices
- Easy cloud expansion

Security

- User authentication
- Role-based authorization
- Encrypted communication
- Secure cloud storage

Performance

- Real-time sensor processing
- Fast AI prediction
- Low response time

Reliability

- Automatic backup
- Fault detection
- Error recovery
- Continuous monitoring

Availability

- 24×7 cloud service
- Load balancing
- High uptime

Maintainability

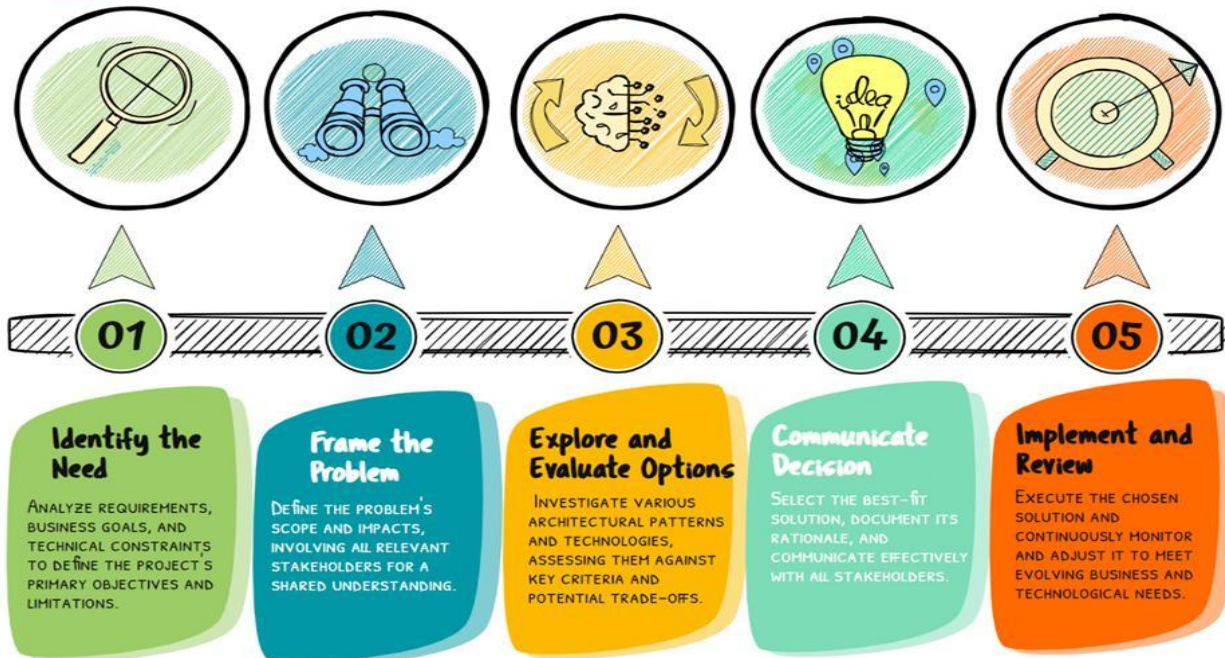
- Modular architecture
- Easy software updates
- Independent modules

6. Architectural Justification :-

The Layered Architecture is the most suitable architecture for the Intelligent AI-Based Smart Dairy Farm Management System because it separates the application into well-defined layers. Each layer performs a specific function and communicates only with adjacent layers, making the system modular, secure, and easy to maintain.

This architecture supports the integration of AI models, IoT devices, cloud services, and databases without affecting other components. It also improves scalability by allowing new services and devices to be added independently. Security is enhanced through a dedicated authentication layer, while maintainability is improved because each layer can be updated or tested separately.

5 architectural decision-making process



Trade-offs

- Slightly higher communication overhead between layers.
- More initial development effort compared to a simple monolithic system.

Future Enhancements

- Blockchain-based milk traceability.
- Drone monitoring for large dairy farms.
- AI chatbot for farmer assistance.
- Automatic veterinary appointment scheduling.
- Predictive maintenance of farm equipment.
- Integration with government agricultural portals.

Conclusion :-

The **Intelligent AI-Based Smart Dairy Farm Management System** is an advanced solution that uses **AI, IoT, wearable sensors, and cloud computing** to improve dairy farm management. It enables continuous cattle health monitoring, disease prediction, feeding optimization, milk production tracking, and real-time farm analytics.

The proposed **Layered Architecture** provides a secure, scalable, reliable, and maintainable system by organizing the application into separate functional layers. This architecture supports efficient communication between components and allows easy future enhancements.

Overall, the system improves livestock welfare, increases dairy productivity, reduces operational costs, and supports data-driven decision-making, making it an effective solution for modern smart dairy farming.

